

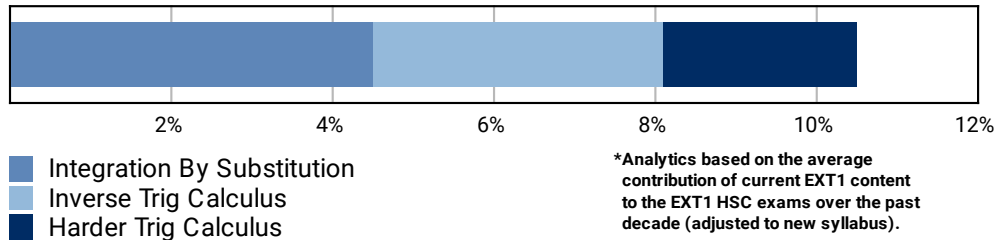
EXT 1: Calculus (Ext1), C2 Further Calculus Skills (Y12)

Inverse Functions Calculus (Ext1)

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Exam Equivalent Time: 93 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

C2 Further Calculus Skills



HISTORICAL CONTRIBUTION

- C2 Further Calculus Skills has a past contribution of 10.5% per HSC exam over the past 10 years.
- This topic has been split into three sub-categories for analysis purposes which are: 1-Integration By Substitution (4.5%), 2-Inverse Trig Calculus (3.6%) and 3-Harder Trig Calculus (2.4%).
- This analysis looks at Inverse Trig Calculus.

HSC ANALYSIS - What to expect and common pitfalls

- Inverse Trig Calculus (3.6%) is regularly tested in the Ext1 exam, appearing in 8 exams over the last decade (including 2 questions in 2021, 2020, 2015, 2013 and 2012).
- The most common question type is standard differentiation or integration - a classic example of low hanging fruit that well-prepared students should expect to score highly in.
- The 2020 exam tweaked this common question type and caused significant problems. 2020 Ext1 13c is a must-review question.
- Examiners have ratcheted up the difficulty in this topic area periodically by requiring students to differentiate inverse trig equations that contain two inverse trig functions and sketch the graph. We recommend students carefully review 2018 Ext1 12c, 2015 Ext1 13d and 2010 Ext1 5b.

Questions

1. Calculus, EXT1 C2 2020 HSC 3 MC

Which of the following is an anti-derivative of $\frac{1}{4x^2 + 1}$?

- A. $2 \tan^{-1}\left(\frac{x}{2}\right) + c$
- B. $\frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) + c$
- C. $2 \tan^{-1}(2x) + c$
- D. $\frac{1}{2} \tan^{-1}(2x) + c$

2. Calculus, EXT1 C2 2012 HSC 9 MC

What is the derivative of $\cos^{-1}(3x)$?

- (A) $\frac{1}{3\sqrt{1-9x^2}}$
- (B) $\frac{-1}{3\sqrt{1-9x^2}}$
- (C) $\frac{3}{\sqrt{1-9x^2}}$
- (D) $\frac{-3}{\sqrt{1-9x^2}}$

3. Calculus, EXT1 C2 2014 HSC 6 MC

What is the derivative of $3 \sin^{-1} \frac{x}{2}$?

- (A) $\frac{6}{\sqrt{4-x^2}}$
- (B) $\frac{3}{\sqrt{4-x^2}}$
- (C) $\frac{3}{2\sqrt{4-x^2}}$
- (D) $\frac{3}{4\sqrt{4-x^2}}$

4. Calculus, EXT1 C2 2015 HSC 7 MC

What is the value of k such that $\int_0^k \frac{1}{\sqrt{4-x^2}} dx = \frac{\pi}{3}$?

- (A) 1
- (B) $\sqrt{3}$
- (C) 2
- (D) $2\sqrt{3}$

5. Calculus, EXT1 C2 2019 HSC 3 MC

What is the derivative of $\tan^{-1} \frac{x}{2}$?

- A. $\frac{1}{2(4+x^2)}$
- B. $\frac{1}{4+x^2}$
- C. $\frac{2}{4+x^2}$
- D. $\frac{4}{4+x^2}$

6. Calculus, EXT1 C2 2006 HSC 1a

Find $\int \frac{dx}{49+x^2}$. (2 marks)

7. Calculus, EXT1 C2 2010 HSC 1a

Use the table of standard integrals to find $\int \frac{1}{\sqrt{4-x^2}} dx$. (1 mark)

8. Calculus, EXT1 C2 2021 HSC 11f

Evaluate $\int_0^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx$. (2 marks)

9. Calculus, EXT1 C2 2004 HSC 1d

Find $\int_0^1 \frac{dx}{\sqrt{4-x^2}}$. (2 marks)

10. Calculus, EXT1 C2 2005 HSC 1a

Find $\int \frac{1}{x^2+49} dx$. (1 mark)

11. Calculus, EXT1 C2 2007 HSC 1c

Differentiate $\tan^{-1}(x^4)$ with respect to x . (2 marks)

12. Calculus, EXT1 C2 2012 HSC 11a

Evaluate $\int_0^3 \frac{1}{9+x^2} dx$. (3 marks)

13. Calculus, EXT1 C2 2013 HSC 11g

Differentiate $x^2 \sin^{-1} 5x$. (2 marks)

14. Calculus, EXT1 C2 2008 HSC 1b

Differentiate $\cos^{-1}(3x)$ with respect to x . (2 marks)

15. Calculus, EXT1 C2 2008 HSC 1c

Evaluate $\int_{-1}^1 \frac{1}{\sqrt{4-x^2}} dx$. (2 marks)

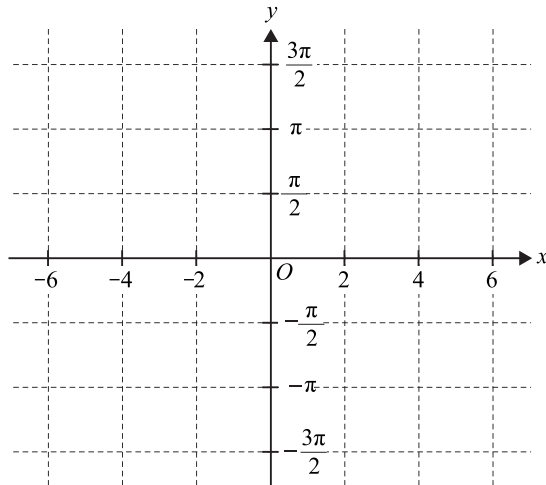
16. Calculus, EXT1 C2 2020 SPEC1 6

Let $f(x) = \tan^{-1}(3x - 6) + \pi$.

a. Show that $f'(x) = \frac{3}{9x^2 - 36x + 37}$. (1 mark)

b. Hence, show that the graph of f has a point of inflection at $x = 2$. (2 marks)

c. Sketch the graph of $y = f(x)$ on the axes provided below. Label any asymptotes with their equations and the point of inflection with its coordinates. (2 marks)



17. Calculus, EXT1 C2 2006 HSC 2a

Let $f(x) = \sin^{-1}(x + 5)$.

i. State the domain and range of the function $f(x)$. (2 marks)

ii. Find the gradient of the graph of $y = f(x)$ at the point where $x = -5$. (2 marks)

iii. Sketch the graph of $y = f(x)$. (2 marks)

18. Calculus, EXT1 C2 2016 HSC 11c

Differentiate $3 \tan^{-1}(2x)$. (2 marks)

19. Calculus, EXT1 C2 2013 HSC 11b

Find $\int \frac{1}{\sqrt{49 - 4x^2}} dx$. (2 marks)

20. Calculus, EXT1 C2 2004 HSC 2b

Find $\frac{d}{dx} \cos^{-1}(3x^2)$. (2 marks)

21. Calculus, EXT1 C2 2005 HSC 2a

Find $\frac{d}{dx} (2 \sin^{-1} 5x)$. (2 marks)

22. Calculus, EXT1 C2 2018 HSC 12c

Let $f(x) = \sin^{-1} x + \cos^{-1} x$.

i. Show that $f'(x) = 0$ (1 mark)

ii. Hence, or otherwise, prove

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}. \quad (1 \text{ mark})$$

iii. Hence, sketch

$$f(x) = \sin^{-1} x + \cos^{-1} x. \quad (1 \text{ mark})$$

23. Calculus, EXT1 C2 2015 HSC 13d

Let $f(x) = \cos^{-1}(x) + \cos^{-1}(-x)$, where $-1 \leq x \leq 1$.

i. By considering the derivative of $f(x)$, prove that $f(x)$ is constant. (2 marks)

ii. Hence deduce that $\cos^{-1}(-x) = \pi - \cos^{-1}(x)$. (1 mark)

24. Calculus, EXT1 C2 2010 HSC 5b

Let $f(x) = \tan^{-1}(x) + \tan^{-1}\left(\frac{1}{x}\right)$ for $x \neq 0$.

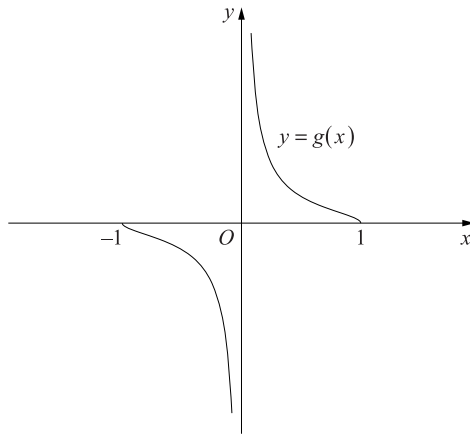
i. By differentiating $f(x)$, or otherwise, show that $f(x) = \frac{\pi}{2}$ for $x > 0$. (3 marks)

ii. Given that $f(x)$ is an odd function, sketch the graph $y = f(x)$. (1 mark)

25. Calculus, EXT1 C2 2020 HSC 13c

Suppose $f(x) = \tan(\cos^{-1}(x))$ and $g(x) = \frac{\sqrt{1-x^2}}{x}$.

The graph of $y = g(x)$ is given.



i. Show that $f'(x) = g'(x)$. (4 marks)

ii. Using part (i), or otherwise, show that $f(x) = g(x)$. (3 marks)

26. Calculus, EXT1 C2 2021 HSC 14e

The polynomial $g(x) = x^3 + 4x - 2$ passes through the point (1, 3).

Find the gradient of the tangent to $f(x) = xg^{-1}(x)$ at the point where $x = 3$. (2 marks)

Worked Solutions

1. Calculus, EXT1 C2 2020 HSC 3 MC

$$\begin{aligned}\int \frac{1}{4x^2 + 1} dx &= \frac{1}{2} \int \frac{2}{1 + (2x)^2} dx \\ &= \frac{1}{2} \tan^{-1}(2x) + c\end{aligned}$$

$\Rightarrow D$

2. Calculus, EXT1 C2 2012 HSC 9 MC

$$y = \cos^{-1}(3x)$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{-1}{\sqrt{1 - (3x)^2}} \times \frac{d}{dx}(3x) \\ &= \frac{-3}{\sqrt{1 - 9x^2}} \\ &\Rightarrow D\end{aligned}$$

3. Calculus, EXT1 C2 2014 HSC 6 MC

$$\begin{aligned}y &= 3 \sin^{-1} \frac{x}{2} \\ \frac{dy}{dx} &= 3 \times \frac{1}{\sqrt{4 - x^2}} \\ &\Rightarrow B\end{aligned}$$

4. Calculus, EXT1 C2 2015 HSC 7 MC

$$\int_0^1 \frac{1}{\sqrt{4-x^2}} dx = \frac{\pi}{3}$$

$$\left[\sin^{-1} \frac{x}{2} \right]_0^k = \frac{\pi}{3}$$

$$\sin^{-1} \frac{k}{2} - \sin^{-1} 0 = \frac{\pi}{3}$$

$$\sin^{-1} \frac{k}{2} = \frac{\pi}{3}$$

$$\frac{k}{2} = \sin \frac{\pi}{3}$$

$$= \frac{\sqrt{3}}{2}$$

$$\therefore k = \sqrt{3}$$

$\Rightarrow B$

5. Calculus, EXT1 C2 2019 HSC 3 MC

$$y = \tan^{-1} \frac{x}{2}$$

$$\frac{dy}{dx} = \frac{\frac{1}{2}}{1 + \left(\frac{x}{2}\right)^2}$$

$$= \frac{1}{2\left(1 + \frac{x^2}{4}\right)}$$

$$= \frac{2}{4 + x^2}$$

$\Rightarrow C$

6. Calculus, EXT1 C2 2006 HSC 1a

$$\int \frac{dx}{49 + x^2} = \frac{1}{7} \tan^{-1} \frac{x}{7} + c$$

7. Calculus, EXT1 C2 2010 HSC 1a

$$\int \frac{1}{\sqrt{4-x^2}} dx = \sin^{-1} \frac{x}{2} + c$$

8. Calculus, EXT1 C2 2021 HSC 11f

$$\begin{aligned} \int_0^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx &= \left[\sin^{-1} \frac{x}{2} \right]_0^{\sqrt{3}} \\ &= \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} 0 \\ &= \frac{\pi}{3} \end{aligned}$$

9. Calculus, EXT1 C2 2004 HSC 1d

$$\begin{aligned} \int_0^1 \frac{dx}{\sqrt{4-x^2}} &= \left[\sin^{-1} \frac{x}{2} \right]_0^1 \\ &= \sin^{-1} \left(\frac{1}{2} \right) - \sin^{-1}(0) \\ &= \frac{\pi}{6} \end{aligned}$$

10. Calculus, EXT1 C2 2005 HSC 1a

$$\begin{aligned} \int \frac{1}{x^2 + 49} dx &= \frac{1}{7} \int \frac{7}{x^2 + 7^2} dx \\ &= \frac{1}{7} \tan^{-1} \frac{x}{7} + c \end{aligned}$$

11. Calculus, EXT1 C2 2007 HSC 1c

$$\begin{aligned}y &= \tan^{-1}(x^4) \\ \frac{dy}{dx} &= \frac{1}{1+(x^4)^2} \times \frac{d}{dx}(x^4) \\ &= \frac{4x^3}{1+x^8}\end{aligned}$$

12. Calculus, EXT1 C2 2012 HSC 11a

$$\begin{aligned}\int_0^3 \frac{1}{9+x^2} dx &= \int_0^3 \frac{1}{3^2+x^2} dx \\ &= \left[\frac{1}{3} \tan^{-1}\left(\frac{x}{3}\right) \right]_0^3 \\ &= \left[\left(\frac{1}{3} \tan^{-1} 1 \right) - 0 \right] \\ &= \frac{1}{3} \times \frac{\pi}{4} \\ &= \frac{\pi}{12}\end{aligned}$$

13. Calculus, EXT1 C2 2013 HSC 11g

$$\begin{aligned}y &= x^2 \sin^{-1} 5x \\ \text{Using the product rule} \\ \frac{dy}{dx} &= x^2 \times \frac{1}{\sqrt{1-(5x)^2}} \times \frac{d}{dx}(5x) + 2x \sin^{-1} 5x \\ &= \frac{5x^2}{\sqrt{1-25x^2}} + 2x \sin^{-1} 5x\end{aligned}$$

14. Calculus, EXT1 C2 2008 HSC 1b

$$\begin{aligned}y &= \cos^{-1}(3x) \\ \frac{dy}{dx} &= - \frac{1}{\sqrt{1-(3x)^2}} \cdot \frac{d}{dx}(3x) \\ &= \frac{-3}{\sqrt{1-9x^2}}\end{aligned}$$

15. Calculus, EXT1 C2 2008 HSC 1c

$$\begin{aligned}\int_{-1}^1 \frac{1}{\sqrt{4-x^2}} dx \\ &= \left[\sin^{-1}\left(\frac{x}{2}\right) \right]_{-1}^1 \\ &= \sin^{-1}\left(\frac{1}{2}\right) - \sin^{-1}\left(-\frac{1}{2}\right) \\ &= \frac{\pi}{6} - \left(-\frac{\pi}{6}\right) \\ &= \frac{\pi}{3}\end{aligned}$$

$$\begin{aligned} \text{a. } f'(x) &= \frac{\frac{d}{dx}(3x-6)}{1+(3x-6)^2} \\ &= \frac{3}{9x^2 - 36x + 37} \end{aligned}$$

$$\text{b. } f''(x) = \frac{3(18x-36)}{(9x^2-36x+37)^2}$$

$$f''(x) = 0 \text{ when } 18x - 36 = 0 \Rightarrow x = 2$$

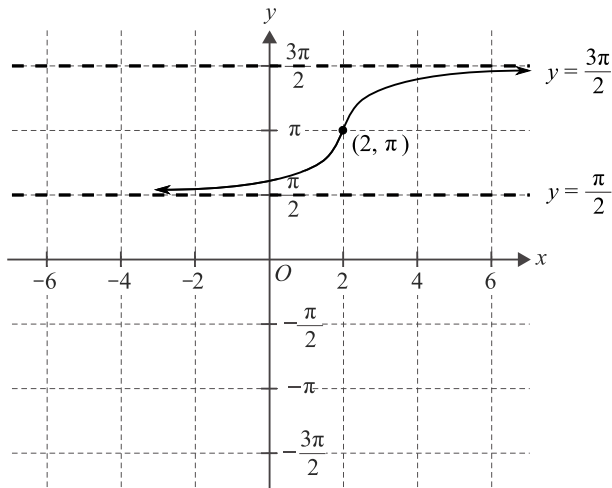
$$\text{If } x < 2, 18x - 36 < 0 \Rightarrow f''(x) < 0$$

$$\text{If } x > 2, 18x - 36 > 0 \Rightarrow f''(x) > 0$$

Since $f''(x)$ changes sign about $x = 2$,

a POI exists at $x = 2$

c.



$$\text{i. } f(x) = \sin^{-1}(x+5)$$

Domain

$$-1 \leq x+5 \leq 1$$

$$-6 \leq x \leq -4$$

Range

$$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

$$\text{ii. } y = \sin^{-1}(x+5)$$

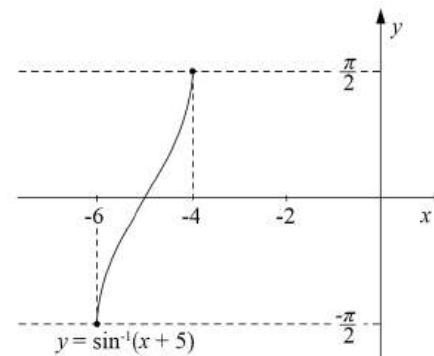
$$\frac{dy}{dx} = \frac{1}{\sqrt{1-(x+5)^2}}$$

When $x = -5$

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{\sqrt{1-(-5+5)^2}} \\ &= \frac{1}{\sqrt{1-0}} \\ &= 1 \end{aligned}$$

$\therefore$ Gradient of $y = f(x)$ at $x = -5$ is 1.

iii.



18. Calculus, EXT1 C2 2016 HSC 11c

$$y = 3 \tan^{-1}(2x)$$

$$\frac{dy}{dx} = \frac{3}{1 + (2x)^2} \times 2$$

$$= \frac{6}{1 + 4x^2}$$

19. Calculus, EXT1 C2 2013 HSC 11b

$$\int \frac{1}{\sqrt{49 - 4x^2}} dx$$

$$= \int \frac{1}{2\sqrt{\frac{49}{4} - x^2}} dx$$

$$= \frac{1}{2} \int \frac{1}{\sqrt{\left(\frac{7}{2}\right)^2 - x^2}} dx$$

$$= \frac{1}{2} \sin^{-1}\left(\frac{2x}{7}\right) + c$$

20. Calculus, EXT1 C2 2004 HSC 2b

$$\frac{d}{dx} \cos^{-1}(3x^2)$$

$$= \frac{-1}{\sqrt{1 - (3x^2)^2}} \times \frac{d}{dx}(3x^2)$$

$$= \frac{-6x}{\sqrt{1 - 9x^4}}$$

21. Calculus, EXT1 C2 2005 HSC 2a

$$\frac{d}{dx}(2 \sin^{-1} 5x) = 2 \times \frac{d}{dx}(\sin^{-1} 5x)$$

$$= 2 \times \frac{1}{\sqrt{1 - (5x)^2}} \times \frac{d}{dx}(5x)$$

$$= \frac{2}{\sqrt{1 - 25x^2}} \times 5$$

$$= \frac{10}{\sqrt{1 - 25x^2}}$$

22. Calculus, EXT1 C2 2018 HSC 12c

i. $f'(x) = \frac{1}{\sqrt{1 - x^2}} + \left(-\frac{1}{\sqrt{1 - x^2}}\right) = 0$

ii. Since $f'(x) = 0 \Rightarrow f(x)$ is a constant.

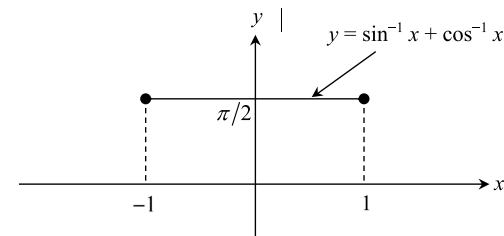
♦ Mean mark 37%.

Substituting $x = 1$ into the equation (any value works)

$$\sin^{-1} 1 + \cos^{-1} 1 = \frac{\pi}{2} + 0$$

$$= \frac{\pi}{2} \dots \text{as required}$$

iii. Domain restrictions require: $-1 < x < 1$



♦ Mean mark 40%.

23. Calculus, EXT1 C2 2015 HSC 13d

i. Prove $f(x)$ is a constant

$$f(x) = \cos^{-1}(x) + \cos^{-1}(-x), \quad -1 \leq x \leq 1$$

$$f'(x) = \frac{-1}{\sqrt{1-x^2}} + \frac{-1}{\sqrt{1-(-x)^2}} \times \frac{d}{dx}(-x)$$

$$= \frac{-1}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-x^2}}$$

$$= 0$$

$\therefore$ Since $f'(x) = 0$, $f(x)$ must be a constant.

ii. $f(0) = \cos^{-1}(0) + \cos^{-1}(0)$

$$= \frac{\pi}{2} + \frac{\pi}{2}$$

$$= \pi$$

$$\therefore f(x) = \pi$$

$$\pi = \cos^{-1}(x) + \cos^{-1}(-x)$$

$$\therefore \cos^{-1}(-x) = \pi - \cos^{-1}(x) \dots \text{as required}$$

♦ Mean mark 35%.

24. Calculus, EXT1 C2 2010 HSC 5b

i. $f(x) = \tan^{-1}(x) + \tan^{-1}\left(\frac{1}{x}\right)$ for $x \neq 0$

$$f'(x) = \frac{1}{1+x^2} + \frac{1}{1+\left(\frac{1}{x}\right)^2} \times \frac{d}{dx}\left(\frac{1}{x}\right)$$

$$= \frac{1}{1+x^2} + \frac{1}{1+\frac{1}{x^2}} \times -\frac{1}{x^2}$$

$$= \frac{1}{1+x^2} - \frac{1}{x^2+1}$$

$$= 0$$

Since $f'(x) = 0$

$\Rightarrow f(x)$ is a constant

Substitute $x = 1$ into $f(x)$

$$f(1) = \tan^{-1} 1 + \tan^{-1}\left(\frac{1}{1}\right)$$

$$= \frac{\pi}{4} + \frac{\pi}{4}$$

$$= \frac{\pi}{2}$$

$$\therefore f(x) = \frac{\pi}{2} \text{ for } x > 0 \dots \text{as required}$$

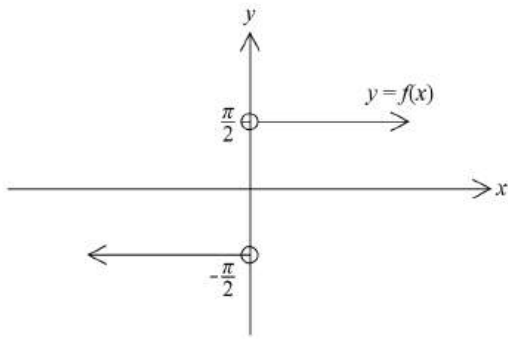
ii. Given $f(x)$ is odd

$$f(-x) = -f(x)$$

♦ Mean mark 48%

MARKER'S COMMENT: The most common errors were not being able to differentiate $\tan^{-1}\left(\frac{1}{x}\right)$ and not recognising the significance of $f'(x) = 0$.

♦ Mean mark 35%



25. Calculus, EXT1 C2 2020 HSC 13c

i. $f(x) = \tan(\cos^{-1}(x))$

$$\begin{aligned} f'(x) &= -\frac{1}{\sqrt{1-x^2}} \cdot \sec^2(\cos^{-1}(x)) \\ &= -\frac{1}{\sqrt{1-x^2}} \cdot \frac{1}{\cos^2(\cos^{-1}(x))} \\ &= -\frac{1}{x^2(1-x^2)} \end{aligned}$$

◆ Mean mark 50%.

$$g(x) = (1-x^2)^{\frac{1}{2}} \cdot x^{-1}$$

$$\begin{aligned} g'(x) &= \frac{1}{2} \cdot -2x(1-x^2)^{-\frac{1}{2}} \cdot x^{-1} - (1-x^2)^{\frac{1}{2}} \cdot x^{-2} \\ &= \frac{-x}{x\sqrt{1-x^2}} - \frac{\sqrt{1-x^2}}{x^2} \\ &= \frac{-x^2 - \sqrt{1-x^2}\sqrt{1-x^2}}{x^2\sqrt{1-x^2}} \\ &= \frac{-x^2 - (1-x^2)}{x^2\sqrt{1-x^2}} \\ &= -\frac{1}{x^2\sqrt{1-x^2}} \\ &= f'(x) \end{aligned}$$

ii. $f'(x) = g'(x)$

$$\Rightarrow f(x) = g(x) + c$$

◆◆ Mean mark 15%.

Find c :

$$f(1) = \tan(\cos^{-1} 1)$$

$$= \tan 0$$

$$= 0$$

$$g(1) = \frac{\sqrt{1-1}}{1} = 0$$

$$f(1) = g(1) + c$$

$$\therefore c = 0$$

$$\therefore f(x) = g(x)$$

26. Calculus, EXT1 C2 2021 HSC 14e

$$g(x) = x^3 + 4x - 2$$

$$g'(x) = 3x^2 + 4$$

$$f(x) = xg^{-1}(x)$$

$$f'(x) = x \cdot \frac{d}{dx}g^{-1}(x) + g^{-1}(x)$$

$g(x)$ passes through $(1, 3)$

$\Rightarrow g^{-1}(x)$ passes through $(3, 1)$

$$g'(1) = 3 + 4 = 7$$

$$\Rightarrow \frac{d}{dx}g^{-1}(3) = \frac{1}{\frac{d}{dy}g(y)} = \frac{1}{g'(1)} = \frac{1}{7}$$

$$\begin{aligned} \therefore f'(x)|_{x=3} &= 3 \cdot \frac{1}{7} + 1 \\ &= \frac{10}{7} \end{aligned}$$

♦♦♦ Mean mark 10%.

COMMENT: The reciprocal relationship of gradients between $g(x)$ and $g^{-1}(x)$ is critical here.